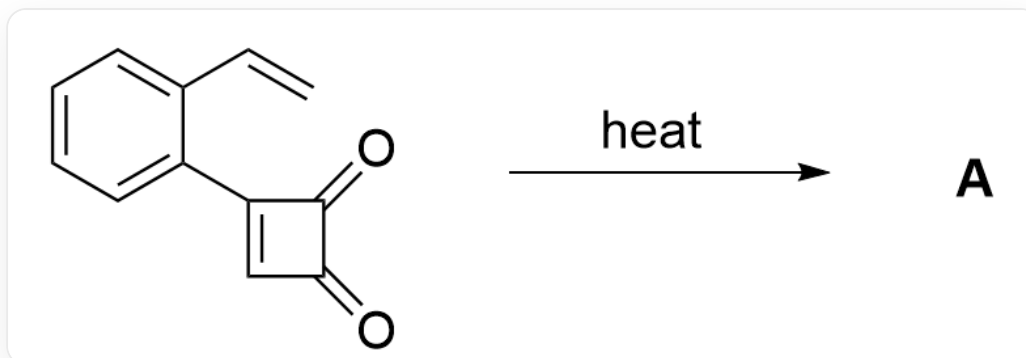


Question

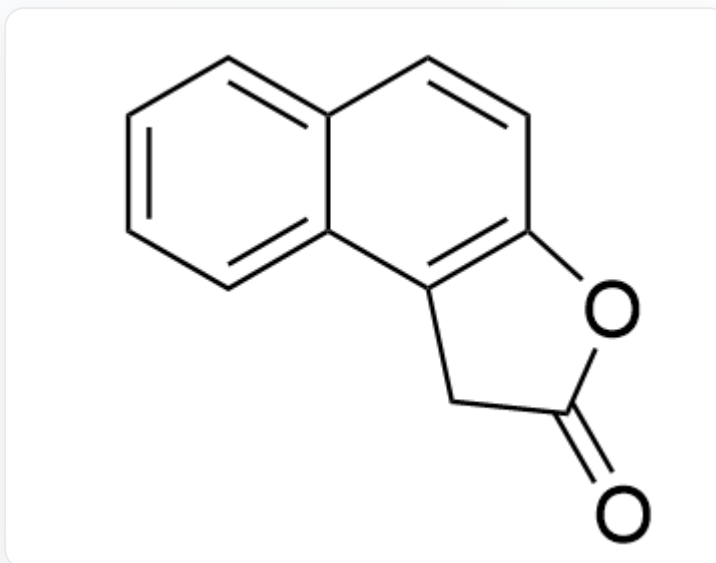


C=CC1=CC=CC=C1C2=CC(C2=O)=O>heat>[A], **A** is the product

The substrate undergoes rearrangement at high temperature to yield product **A**. Given that the molecular formula of **A** is $C_{12}H_8O_2$, provide the structural formula of **A**.

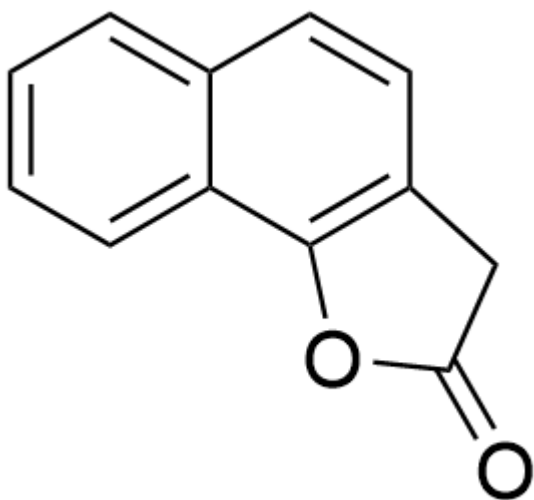
A. All other options are incorrect

B.



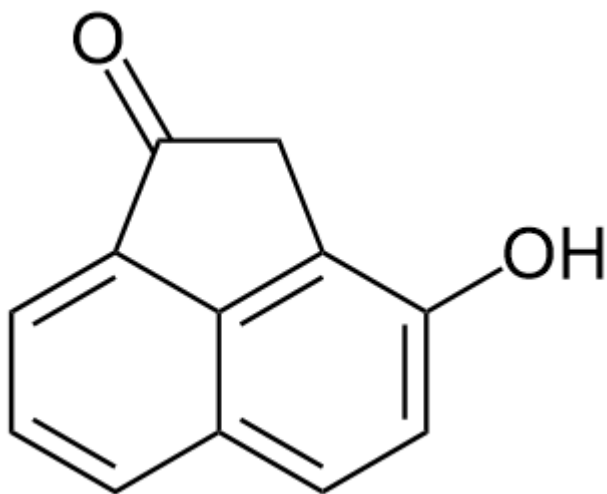
O=C(O1)CC2=C1C=CC3=CC=CC=C32

C.



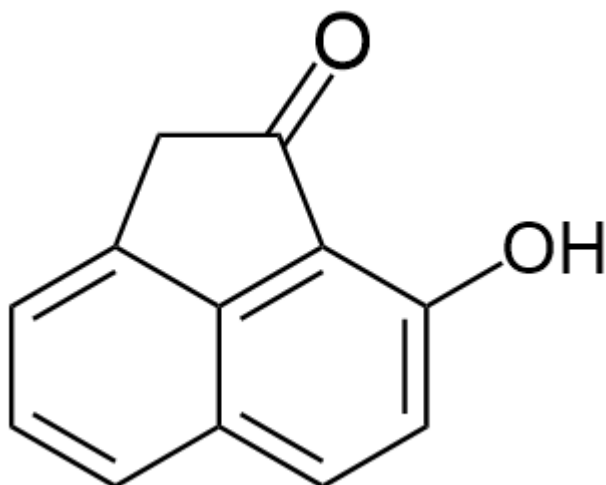
O=C(C1)OC2=C1C=CC3=CC=CC=C32

D.



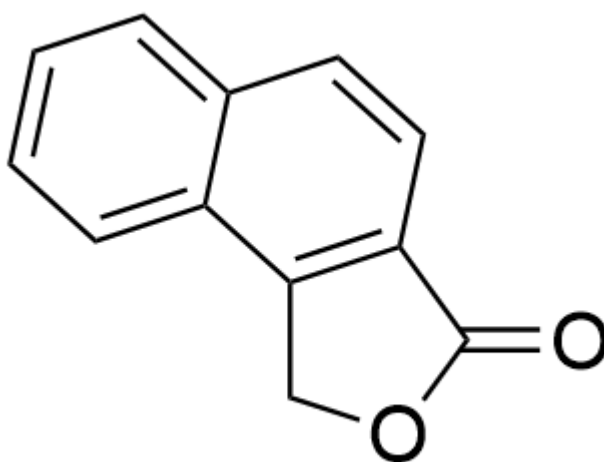
OC1=CC=C2C(C3=CC=CC=C3)C(=O)CC2=O

E.



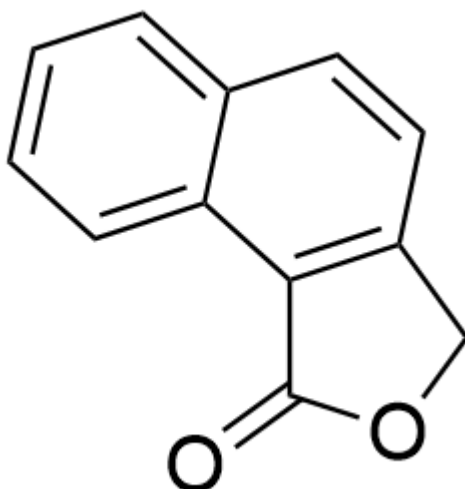
OC1=CC=C2C(C3=CC=C2)=C1C(C3)=O

F.



O=C1C2=C(CO1)C3=CC=CC=C3C=C2

G.



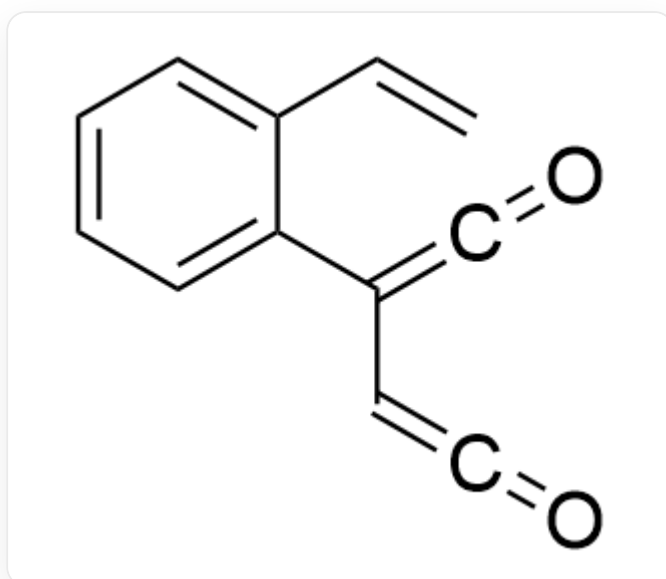
O=C1C(C2=CC=CC=C2C=C3)C=CCO1

Answer

Correct Answer: **B**

Detailed Explanation

First, under the action of high temperature, the reaction substrate undergoes a 4-membered ring-opening reaction to obtain intermediate **1**



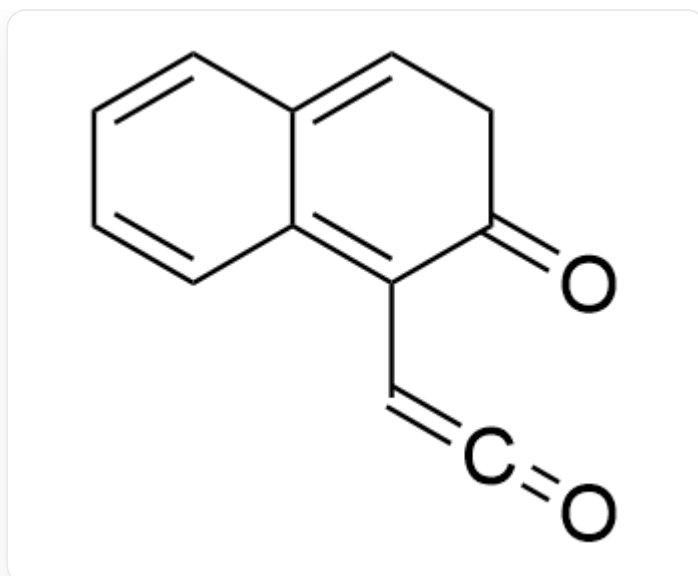
intermediate**1** : C=CC1=CC=CC=C1C(C=O)=C=O

CHECKPOINT

1 PTS

intermediate **1** : C=CC1=CC=CC=C1C(C=O)=C=O

Next, intermediate **1** undergoes an intramolecular 6 π -electron electrocyclic reaction to obtain intermediate **2**



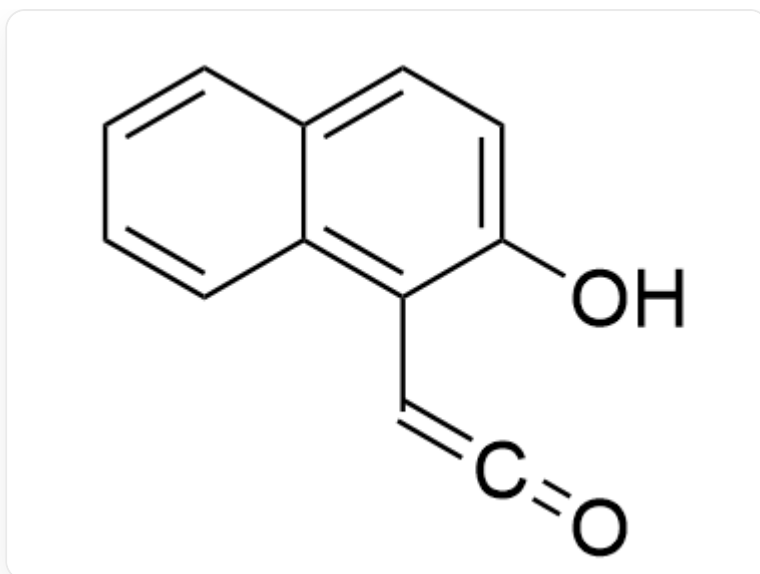
intermediate **2** : O=C1C(C=C=O)=C2C=CC=CC2=CC1

CHECKPOINT

1 PTS

intermediate **2** : O=C1C(C=C=O)=C2C=CC=CC2=CC1

Intermediate **2** undergoes tautomerization to obtain the aromatic intermediate **3**



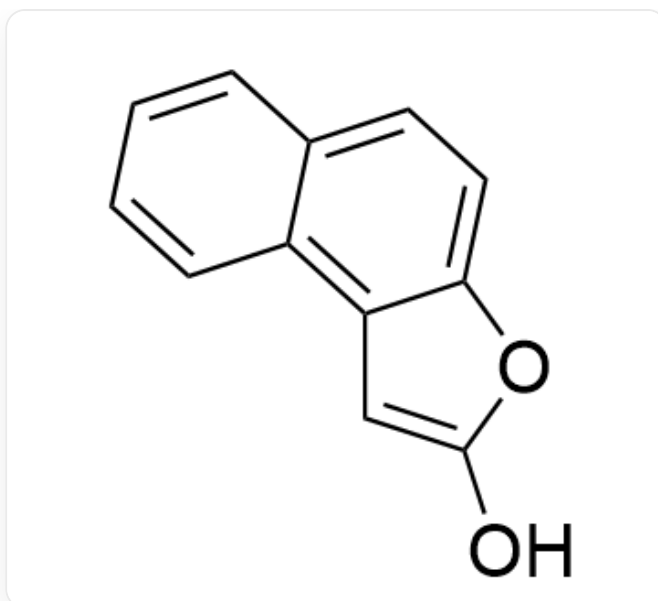
intermediate **3** : OC1=CC=C2C(C=CC=C2)=C1C=C=O

CHECKPOINT

1 PTS

intermediate **3** : OC1=CC=C2C(C=CC=C2)=C1C=C=O

Intermediate **3** still contains an unstable ketene structure, so the hydroxyl group further undergoes nucleophilic cyclization to obtain intermediate **4**



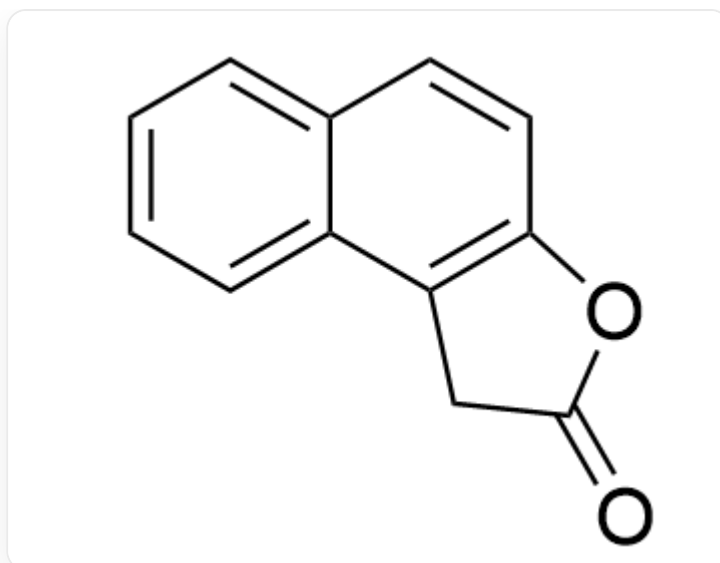
intermediate4 : OC1=CC2=C3C=CC=CC3=CC=C2O1

CHECKPOINT

1 PTS

intermediate 4 : OC1=CC2=C3C=CC=CC3=CC=C2O1

Intermediate 4 undergoes tautomerization to obtain the final product A



product **A** : O=C(O1)CC2=C1C=CC3=CC=CC=C32